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H-MIDE®

LF-0211

Composition : Each 2 mL ampoule contains **Furosemide 20 mg**

Indications :
It is indicated to all forms of cardiac edema that cannot be eliminated by adequate glycoside treatment. Acute cardiac insufficiency, especially in pulmonary edema (administration in conjunction with other therapeutic measures). Reduced urinary output due to gestoses (after restoring the fluid volume to normal. Hypertensive crisis (in addition to other antihypertensive measures). Ascites due to cirrhosis of the liver, mechanical obstruction or cardiac failure. Edema of renal origin. Edema due to burns. Edema of pregnancy after the second trimester, Cerebral Edema, Poisoning, Peripheral edema.

Dosage And Administration :
In general, the dose used must be the lowest which is sufficient to achieve the desired effect. The usual adult intramuscular or intravenous dose is 20-40 mg given as a single injection. The intravenous administration of Furosemide is indicated when a rapid onset of diuresis is desired, e.g., in acute pulmonary edema.
If gastrointestinal absorption is impaired or oral medication is not practical for any reason, Furosemide is indicated by the intravenous. Parenteral use should be replaced with oral Furosemide as soon as practical. Intramuscular administration must be restricted to exceptional cases where neither oral nor intravenous administrations are feasible. Intramuscular injection is not suitable for the treatment of acute conditions such as pulmonary edema.
Give the IV injection slowly (1-2 minutes). If necessary, after not less than two hours, increase by 20 mg until desired diuretic response is obtained. There after, once or twice daily or as directed by the physician.

Acute pulmonary edema- The usual initial dose is 40 mg IV (over 1 to 2 minutes). If the patients condition requires it, a further dose of 20-40 mg is injected after 20 minutes. Additional therapy (eg, digitalis, oxygen) may be given concomitantly.
Forced Diuresis-20-40 mg is given in addition to infusion of electrolyte solution. Further treatment depends on the elimination of urine and must include substitution of the fluid and electrolyte losses.
In poisoning with acid or basic substances, the elimination rate can be further increased by alkalinization or acidification, respectively, of the urine.
Infants and children under 15 years - In principle, Furosemide should be administered orally. Parenteral administration if necessary, (continuous drip infusion) is indicated only in life-threatening conditions. The dosage schedule is 1 mg furosemide per kg body weight up to a daily maximum of 20 mg.

Incompatibilities
pH control during dilution, mixing with other drug and IV incompatibility-Furosemide Injection is buffered alkaline solution with a pH of about 9 and the drug may precipitate at pH values below 7. Care must be taken to ensure that the pH of the prepared infusion solution is in the weakly alkaline to neutral range. Acid solutions, including other parenteral medications (e.g., labetalol, ciprofloxacin, amrinone, and milrinone) must not be administered concurrently in the same infusion because they may cause precipitation of the Furosemide. In addition, Furosemide injection should not be added to a running intravenous line containing any of these acidic products.
If the physician elects to use high dose parenteral therapy, add the Furosemide to either sodium chloride injection, lactated ringers injection, or dextrose (5%) injection after pH has been adjusted to above 5.5, and administer as a controlled intravenous infusion at a rate not greater than 4 mg / min.
A precipitate formed when Furosemide was admixed with gentamicin, netilmicin or mitrinone in 5% Dextrose or 0.9% Sodium Chloride, but not with amikacin, kanamycin or tobramycin. Furosemide admixed with cefoperazone sodium in 5% Dextrose is stable for 2 days at 25°C (77°F) and 5 days at 4°C (39°F)

Pharmacology Pharmacokinetics :
Absorption : In one study in patients with normal renal function, approximately 60% of a single 80-mg oral dose of furosemide was absorbed from the GI tract. When administered to fasting adults in this dosage, the drug appeared in the serum within 10 minutes, reached a peak concentration of 2.3 µg / mL in 60-70 minutes, and was almost completely cleared from the serum in 4 hours.
When the same dose was given after a meal, the serum concentration of furosemide increased slowly to a peak of about 1 µg / mL after 2 hours and similar concentrations were present 4 hours after ingestion. However, a similar diuretic response occurred regardless of whether the drug was given with food or to fasting patients.
The maximum hypotensive effect may not be apparent until several days after furosemide therapy is begun. After IV administration of furosemide, diuresis occurs within 5 minutes, reaches a maximum within 20-60 minutes, and persists for approximately 2 hours. After IM administration, peak plasma concentrations are attained within 30 minutes; onset of diuresis occurs somewhat later than after IV administration. In patients with severely impaired renal function, the diuretic response may be prolonged.
Distribution : Only limited information is available on the distribution of furosemide. The drug crosses the placenta and is distributed into milk.

Furosemide is approximately 95% bound to plasma proteins in both normal and azotemic patients.
Elimination : Plasma concentrations of furosemide decline in a biphasic manner. Various investigators have reported a wide range of elimination half-life for furosemide. In one study, the elimination half-life averaged about 30 minutes in healthy patients who received 20-120 mg of the drug IV.
In another study, the elimination half-life averaged 9.7 hours in patients with advanced renal failure who received 1 g of furosemide IV. The elimination half-life was more prolonged in 1 patient with concomitant liver disease.
In patients with normal renal function, a small amount of furosemide is metabolized in the liver to the defurfurylated derivative, 4-chloro-5-sulfamoylanthranilic acid. Furosemide and its metabolite are rapidly excreted in urine by glomerular filtration and by secretion from the proximal tubule. In patients with normal renal function, approximately 80% of an IV or IM dose are excreted in urine within 24 hours; 69-97% of these amounts is excreted in the first 4 hours. The remainder of the drug is eliminated by nonrenal mechanisms including degradation in the liver and excretion of unchanged drug in the feces. In patients with marked renal impairment without liver disease, nonrenal clearance of furosemide is increased so that up to 98% of the drug is removed from the plasma within 24 hours. One patient with uremia and hepatic cirrhosis eliminated only 58% of an IV dose in 24 hours. Furosemide is not removed by hemodialysis.

Side Effects Adverse Reactions :
Fluid and Electrolyte Disturbances
Furosemide may produce profound diuresis resulting in fluid and electrolyte depletion. Fluid and electrolyte depletion are especially likely to occur when large doses are given and / or in patients with restricted sodium intake.
Excessive dehydration is most likely to occur in geriatric patients and / or patients with chronic cardiac disease treated with prolonged sodium restriction or those receiving sympatholytic agents.
Potassium depletion occurs frequently in patients with secondary hyperaldosteronism which may be associated with cirrhosis or nephrosis and is particularly important in cirrhotic, nephrotic, or digitalized patients.
Adverse reactions are categorized below by organ system and listed by decreasing severity.

- Gastrointestinal System Reactions**
1. Pancreatitis 2. Jaundice (intrahepatic cholestatic jaundice) 3. Anorexia 4. Oral and gastric irritation 5. Cramping 6. Diarrhea 7. Constipation 8. Nausea 9. Vomiting
Systematic Hypersensitivity Reactions
1. Systemic vasculitis 2. Interstitial nephritis 3. Necrotizing angitis
Central Nervous System Reactions
1. Tinnitus and hearing loss 2. Paresthesias 3. Vertigo 4. Dizziness 5. Headache 6. Blurred vision 7. Xanthopsia
Hematologic Reactions
1. Aplastic anemia (rare) 2. Thrombocytopenia 3. Agranulocytosis (rare)
4. Hemolytic anemia 5. Leukopenia 6. Anemia
Dermatologic Reactions
1. Exfoliative dermatitis 2. Erythema multiforme 3. Purpura 4. Photosensitivity 5. Urticaria 6. Rash 7. Pruritus
Cardiovascular Reaction
Orthostatic hypotension may occur and be aggravated by alcohol, barbiturates or narcotics.
Other Reactions
1. Hyperglycemia 2. Glycosuria 3. Hyperuricemia 4. Muscle spasm 5. Weakness 6. Restlessness 7. Urinary bladder spasm 8. Thrombophlebitis 9. Fever
Whenever adverse reactions are moderate or severe, Furosemide dosage should be reduced or therapy withdrawn.

Precautions :
General : Excessive diuresis may cause dehydration and blood volume reduction with circulatory collapse and possibly vascular thrombosis and embolism, particularly in elderly patients. As with any effective diuretic, electrolyte depletion may occur during Furosemide therapy, especially in patients receiving higher doses and a restricted salt intake.
Hypokalemia may develop with Furosemide, especially with brisk diuresis, inadequate oral electrolyte intake, when cirrhosis is present, or during concomitant use of corticosteroids or ACTH. Digitalis therapy may exaggerate metabolic effects of hypokalemia, especially myocardial effects.
All patients receiving Furosemide therapy should be observed for these signs or symptoms of fluid or electrolyte imbalance (hyponatremia, hypochloremic alkalosis, hypokalemia, hypomagnesemia or hypocalcemia) : dryness of mouth, thirst, weakness, lethargy, drowsiness, restlessness, muscle pains or cramps, muscular fatigue, hypotension, oliguria, tachycardia, arrhythmia, or gastrointestinal disturbances such as nausea and vomiting.
Increases in blood glucose and alterations in glucose tolerance tests (with abnormalities of the fasting and 2-hour postprandial sugar) have been observed, and rarely, precipitation of

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diabetes mellitus had been reported.
Asymptomatic hyperuricemia can occur and gout may rarely be precipitated.
Patients allergic to sulfonamides may also be allergic to Furosemide. The possibility exists of exacerbation or activation of systemic lupus erythematosus.
As with many other drugs, patients should be observed regularly for the possible occurrence of blood dyscrasias, liver or kidney damage, or other idiosyncratic reactions.
Pregnancy : Pregnancy Category C : Furosemide has been shown to cause unexplained maternal deaths and abortions in rabbits at 2, 4, , and 8 times the maximal recommended human dose.
There are no adequate and well-controlled studies in pregnant women. Furosemide should be used during pregnancy only if the potential benefit justifies the potential risk to the fetus.
The effects of Furosemide on embryonic and fetal development and on pregnant dams were studied in mice, rats and rabbits.
Furosemide caused unexplained maternal deaths and abortions in the rabbit at the lowest dose of 25 mg / kg (2 times the maximal recommended human dose of 600 mg / day). In another study, a dose of 50 mg / day (4 times the maximal recommended human dose of 600 mg / day) also caused maternal deaths and abortions when administered to rabbits between Days 12 and 17 of gestation. In a third study, none of the pregnant rabbits survived a dose of 100 mg / kg. Data from the above studies indicated fetal lethality that can precede maternal deaths.
The results of the mouse study and one of the three rabbit studies also showed an increased incidence and severity of hydronephrosis (distention of the renal pelvis and, in some cases, of the ureters) in fetuses derived from treated dams as compared to the incidence in fetuses from the control group.

Nursing Mothers : Because it appears in breast milk, caution should be exercised when Furosemide is administered to a nursing mother.
Drug Interactions : Furosemide may increase the ototoxic potential of aminoglycoside antibiotics, especially in the presence of impaired renal function. Except in life threatening situations, avoid this combination. It has been proposed, but not proven, that furosemide may enhance the nephrotoxicity of neomycin.
Furosemide should not be used concomitantly with ethacrynic acid because of the possibility of ototoxicity. Patients receiving high doses of salicylates concomitantly with Furosemide, as in rheumatic disease, may experience salicylate toxicity at lower doses because of competitive renal excretory sites. Additive ototoxicity may occur with concomitant use of Cisplatin.
Furosemide has a tendency to antagonize the skeletal muscle relaxing effect of tubocurarine and may potentiate the action of succinylcholine.
Some drugs such as corticosteroids, corticotropin, and amphotericin B also cause potassium loss, and severe potassium depletion may occur when one of these drugs is administered during furosemide therapy. The hypokalaemia associated with diuretics may be exacerbated by carbene xolone, large amount of liquorice and laxatives.

Diuretic-induced electrolyte disturbances may predispose to digitalis-induced arrhythmias. Drugs inducing Q-T interval may have its half-life prolonged in patients who are concurrently taking drugs that can alkalinize the urine such as Furosemide.
Lithium generally should not be given with diuretics because they reduce lithium's renal clearance and add a high risk of lithium toxicity.
Furosemide may add to or potentiate the therapeutic effect of other antihypertensive drugs or ACE inhibitors. Potentiation occurs with ganglionic or peripheral adrenergic blocking drugs. Dosage of the hypotensive agent, and possibly both drugs, should be reduced when furosemide is added to an existing regimen.
Furosemide may decrease arterial responsiveness to norepinephrine. However, norepinephrine may still be used effectively. Simultaneous administration of sucralfate and Furosemide tablets may reduce the nutrieutic and antihypertensive effects of Furosemide. Patients receiving both drugs should be observed closely to determine if the desired diuretic and / or antihypertensive effect of Furosemide is achieved. The intake of Furosemide and sucralfate should be separated by at least two hours.
Administration of furosemide to diabetic patients may interfere with the hypoglycemic effect of insulin or oral antidiabetic agents, possibly as a result of hypokalemia. Patients should be observed for possible decrease of diabetic control. If correction of the potassium deficit does not restore control, dosage adjustments of the antidiabetic agent may be needed.

One study in six subjects demonstrated that the combination of Furosemide and acetylsalicylic acid temporarily reduced creatinine clearance in patients with chronic renal insufficiency. There are case reports of patients who developed increased BUN, serum creatinine and serum potassium levels, and weight gain when Furosemide was used in conjunction with NSAIDs.
Literature reports indicate that coadministration of indomethacin may reduce the natriuretic and antihypertensive effects of Furosemide in some patients by inhibiting prostaglandin synthesis. Indomethacin may also affect plasma renin levels, aldosterone excretion, and renin profile evaluation. Patients receiving both indomethacin and Furosemide should be observed closely to determine if the desired diuretic and / or antihypertensive effect of Furosemide is achieved.
In one study, epileptic patients receiving chronic anticonvulsant therapy had a reduced diuretic response to furosemide as compared to controls. All of the epileptic patients were receiving phenytoin sodium and phenobarbital and some were also receiving other anticonvulsants. It has been postulated that renal sensitivity to furosemide is diminished by these drugs.
A reaction characterized by diaphoresis, flushes, variable blood pressure including hypertension, and uneasiness has been reported in some patients with acute myocardial infarction and congestive heart failure who received furosemide IV within 24 hours after administration of an oral hypnotic dose of chloral hydrate.

It has been suggested that furosemide, by increasing serum uric acid concentrations, may interfere with the uricosuric effects of probenecid or sulfinpyrazone. Serum uric acid concentrations should be monitored in patients receiving both drugs, and dosage of the uricosuric drug should be increased if necessary.
Warnings : In patients with hepatic cirrhosis and ascites, Furosemide therapy is best initiated in the hospital. In hepatic coma and in states of electrolyte depletion, therapy should not be instituted until the basic condition is improved. Sudden alterations of fluid and electrolyte balance in patients with cirrhosis may precipitate hepatic coma; therefore, strict observation is necessary during the period of diuresis. Supplemental potassium chloride and, if required, an aldosterone antagonist are helpful in preventing hypokalemia and metabolic alkalosis.
If increasing azotemia and oliguria occur during treatment of severe progressive renal disease, Furosemide should be discontinued.
Urine output must be secured, in those patient whose outflow is obstructed careful monitoring should be done at the beginning of the treatment. Particularly careful surveillance is necessary in Hypoproteinaemia, Renal failure, Premature infants (nephrolithiasis and nephrocalcinosis), Gout, and Hypotension.
Cases of tinnitus and reversible or irreversible hearing impairment have been reported. Usually, reports indicate that Furosemide ototoxicity is associated with rapid injection, severe renal impairment, doses exceeding several times the usual recommended dose, or concomitant therapy with aminoglycoside antibiotics, ethacrynic acid, or other ototoxic drugs. If the physician elects to use high dose parenteral therapy, controlled intravenous infusion is advisable (for adults, an infusion rate not exceeding 4 mg Furosemide per minute has been used).
Monitoring : During treatment with Furosemide, serum sodium, potassium and creatinine should be monitored regularly. Patients at risk of developing electrolyte imbalances, and those with significant additional fluid loss due e.g. vomiting, diarrhea or intense sweating, must be closely monitored. Hypovolaemia or dehydration, as well as any significant disturbances in electrolyte content and acid-base balance, must be corrected. A temporary discontinuation of treatment with Furosemide may become necessary.

Contraindications :
Furosemide is contraindicated in patients with the following :
Anuria; Hepatic coma and precoma; Severely reduced blood levels of potassium (hypokalaemia) or of sodium (hyponatraemia); Hypovolaemia with or without hypotension or dehydration; Patients allergic to sulphonamides may show cross-sensitivity to furosemide; Lactation; Pregnancy unless there are compelling medical reasons. If given, fetal growth must be monitored; and in patients with a history of hypersensitivity to Furosemide or any of the excipients.

Symptoms and Treatment for Overdosage and Antidote :
The principal signs and symptoms of overdose with Furosemide are dehydration, blood volume reduction, hypotension, electrolyte imbalance, hypokalemia and hypochloremic alkalosis, and are extension of its diuretic action.
Treatment of overdose is supportive and consists of replacement of excessive fluid and electrolyte losses. Serum electrolytes, carbon dioxide level and blood pressure should be determined frequently. Adequate drainage must be assured in patients with urinary bladder outlet obstruction (such as prostatic hypertrophy).
Hemodialysis does not accelerate Furosemide elimination.

Storage Conditions :
Store below 30 °C. Protect from light.

Shelf-life :
4 years

Product Description :
Clear, colorless sterile solution for injection.

Pack Size Presentation :
1 x [50 x 2 mL / amp.]

Registration Number :
MAL20032248AZ

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FRONT

BACK

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LF F&B

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